

CO3 - ASSESSMENT TOOL -1

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Course Name : Ethical Hacking

Question 1

A cyber security consulting company has been contracted to perform an initial security assessment of the publicly accessible server scanme.nmap.org. Using Nmap, perform host discovery, TCP port scanning, service version detection, operating system fingerprinting, and default script scanning on scanme.nmap.org.

KaliLinux Commands

1. Host Discovery

```
nmap -sn scanme.nmap.org
```

2. TCP Port Scanning

```
nmap -sS scanme.nmap.org
```

3. Service Version Detection

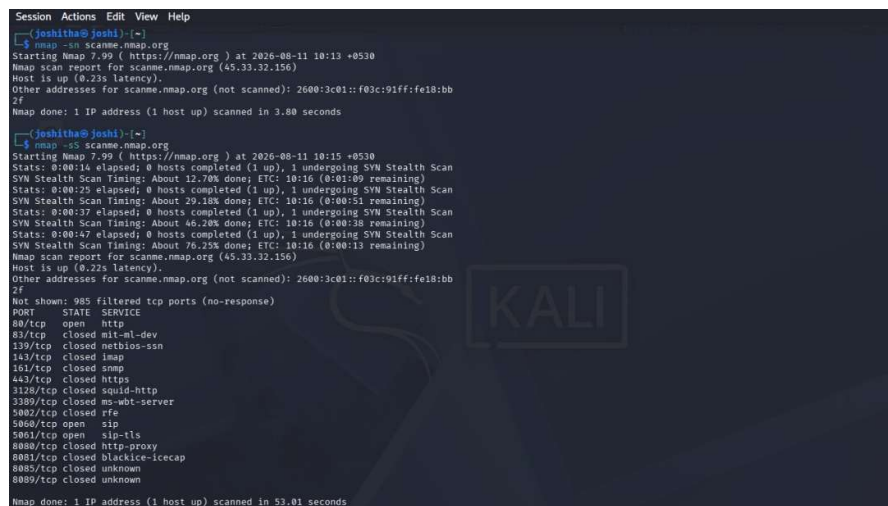
```
nmap -sV scanme.nmap.org
```

4. Operating System Fingerprinting

```
sudo nmap -O scanme.nmap.org
```

5. Default Script Scanning

```
nmap -sC scanme.nmap.org
```



```
Session Actions Edit View Help
[joshitha@joshi]~$ nmap -sn scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:13 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.23s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb
2f
Nmap done: 1 IP address (1 host up) scanned in 3.80 seconds

[joshitha@joshi]~$ nmap -sS scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:15 +0530
Stats: 0:00:16 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 12.7% done; ETC: 10:16 (0:01:09 remaining)
Stats: 0:00:25 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 29.18% done; ETC: 10:16 (0:00:51 remaining)
Stats: 0:00:37 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 46.28% done; ETC: 10:16 (0:00:38 remaining)
Stats: 0:00:47 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 76.25% done; ETC: 10:16 (0:00:13 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.22s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb
2f
Not shown: 985 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
81/tcp    closed ntlm-dev
139/tcp   closed netbios-ssn
143/tcp   closed imap
161/tcp   closed snmp
443/tcp   closed https
3128/tcp  closed squid-http
3389/tcp  closed ms-wbt-server
5802/tcp  closed rfb
5850/tcp  open  sip
5851/tcp  open  sip-tls
8080/tcp  closed http-proxy
8081/tcp  closed blackice-icecap
8085/tcp  closed unknown
8089/tcp  closed unknown
Nmap done: 1 IP address (1 host up) scanned in 53.01 seconds
```

```

Session Actions Edit View Help
l-$ nmap -sV scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:16 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.21s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb
2f
Not shown: 985 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
83/tcp    closed mit-ml-dev
139/tcp   closed netbios-ssn
143/tcp   closed imap
161/tcp   closed snmp
443/tcp   closed https
3128/tcp  closed squid-http
3389/tcp  closed ms-wbt-server
5002/tcp  closed rfe
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
8080/tcp  closed http-proxy
8081/tcp  closed blackice-icecap
8085/tcp  closed unknown
8089/tcp  closed unknown

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 71.19 seconds

```

```

Session Actions Edit View Help
443/tcp   closed https
3128/tcp  closed squid-http
3389/tcp  closed ms-wbt-server
5002/tcp  closed rfe
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
8080/tcp  closed http-proxy
8081/tcp  closed blackice-icecap
8085/tcp  closed unknown
8089/tcp  closed unknown
Device type: general purpose firewall/router
Running (JUST GUESSING): Linux 4.X|6.X|2.6.X|3.X|5.X (97%), IPFire 2.X (91%), Mikrotik RouterOS 7.X (87%)
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:6.1 cpe:/o:linux:linux_kernel:2.6.32 cpe:/o:linux:linux_kernel:3.10 cpe:/o:linux:linux_kernel:5.4 cpe:/o:ipfire:ipfire:2.27 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:2.6.32
Aggressive OS guesses: Linux 4.19 - 5.15 (97%), Linux 6.1 (94%), Linux 2.6.32 (92%), Linux 2.6.32 or 3.10 (92%), Linux 4.0 - 4.4 (92%), Linux 4.15 (92%), Linux 5.4 (91%), IPFire 2.27 (Linux 5.15 - 6.1) (91%), Linux 6.12 (91%), Linux 2.6.32 - 2.6.35 (89%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 17 hops

TRACEROUTE (using port 3389/tcp)
HOP RTT ADDRESS
1 2.97 ms 172.25.16.1
2 2.38 ms 172.22.10.1
3 6.01 ms 14.139.187.225
4 142.21 ms 103.249.82.241
5 142.52 ms 61.95.248.189
6 142.82 ms 116.119.36.60
7 143.10 ms 182.79.198.0
8 143.41 ms 116.119.42.7
9 ...
10 240.96 ms unknown.telstraglobal.net (202.84.207.181)
11 395.81 ms 1-1120-egnx-core02.telstraglobal.net (202.84.141.153)
12 395.30 ms 1-92-egnx03.telstraglobal.net (202.84.247.17)
13 394.96 ms eqix-sv1.linode.com (206.223.116.196)
14 264.87 ms 10.203.32.4
15 ...
16 253.00 ms 10.203.7.26
17 252.58 ms scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 127.73 seconds

```

```

Session Actions Edit View Help
s://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 127.73 seconds

(joshitha@joshi)-[~]
l-$ nmap -sC scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:23 +0530
Stats: 0:00:44 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 70.17% done; ETC: 10:24 (0:00:16 remaining)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.33s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb
2f
Not shown: 985 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
83/tcp    closed mit-ml-dev
139/tcp   closed netbios-ssn
143/tcp   closed imap
161/tcp   closed snmp
443/tcp   closed https
3128/tcp  closed squid-http
3389/tcp  closed ms-wbt-server
5002/tcp  closed rfe
5060/tcp  open  sip
5061/tcp  open  sip-tls
8080/tcp  closed http-proxy
8081/tcp  closed blackice-icecap
8085/tcp  closed unknown
8089/tcp  closed unknown

Nmap done: 1 IP address (1 host up) scanned in 82.84 seconds

(joshitha@joshi)-[~]
l-$

```

Question 2

The security team has deployed a deliberately vulnerable Linux server, 192.168.56.101 (Metasploitable 2), in an isolated laboratory environment to train penetration testers. As part of the pre-engagement reconnaissance phase, use Nmap to perform comprehensive enumeration of 192.168.56.101.

Kali Linux Commands

```
nmap -sn 192.168.56.101
```

```
nmap -Pn -sS 192.168.56.101
```

```
nmap -Pn -sV 192.168.56.101
```

```
sudo nmap -Pn -O 192.168.56.101
```

```
sudo nmap -Pn -A 192.168.56.101
```

```
(joshitha@joshi):[~]
$ nmap -sn 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:28 +0530
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.02 seconds
```

```
(joshitha@joshi):[~]
$ nmap -Pn -sS 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:32 +0530
Stats: 0:00:21 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 8.50% done; ETC: 10:36 (0:03:14 remaining)
Stats: 0:01:59 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 41.60% done; ETC: 10:37 (0:02:43 remaining)
Stats: 0:06:34 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 66.27% done; ETC: 10:42 (0:03:19 remaining)
Stats: 0:08:47 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 70.40% done; ETC: 10:45 (0:03:40 remaining)
Stats: 0:10:40 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 73.80% done; ETC: 10:47 (0:03:46 remaining)
Stats: 0:11:39 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 75.60% done; ETC: 10:48 (0:03:45 remaining)
Stats: 0:16:28 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 84.30% done; ETC: 10:52 (0:03:03 remaining)
Stats: 0:18:36 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 88.13% done; ETC: 10:53 (0:02:30 remaining)
Stats: 0:19:35 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 89.93% done; ETC: 10:54 (0:02:11 remaining)
Stats: 0:22:15 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 94.73% done; ETC: 10:56 (0:01:14 remaining)
Stats: 0:24:32 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 98.90% done; ETC: 10:57 (0:00:16 remaining)
Stats: 0:24:45 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 99.30% done; ETC: 10:57 (0:00:10 remaining)
Nmap scan report for 192.168.56.101
Host is up (0.041s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE
1074/tcp  closed warmspotMgmt
2000/tcp  closed cisco-sccp
4000/tcp  closed remoteanything
5060/tcp  open   sip
5061/tcp  open   sip-tls
Nmap done: 1 IP address (1 host up) scanned in 1510.48 seconds
```

```
(joshitha@joshi):[~]
$ nmap -Pn -sV 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:24 +0530
Nmap scan report for 192.168.56.101
Host is up (0.00063s latency).
Not shown: 981 closed tcp ports (conn-refused)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.2p4 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rcpbind      2.4.0 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
3306/tcp   open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp   open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp   open  vnc           VNC (protocol 3.3)
Service detection performed. Please report any incorrect results at https://nmap.org/subbit/ .
Nmap done: 1 IP address (1 host up) scanned in 54.21 seconds
```

```

[joshitha@josh]~$ sudo nmap -Pn -O 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:26 +0530
Nmap scan report for 192.168.56.101
Host is up (0.00056s latency).
Not shown: 981 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
111/tcp   open  rpcbind
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:6B:3F:8D (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 18.73 seconds

```

```

[joshitha@josh]~$ sudo nmap -Pn -A 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:28 +0530
Nmap scan report for 192.168.56.101
Host is up (0.00050s latency).
Not shown: 981 closed tcp ports (conn-refused)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2-4 (RPC #100000)
445/tcp   open  smb          Samba smbd 3.X - 4.X (workgroup: WORKGROUP)

Host script results:
|_ smb-os-discovery:
|_   OS: Unix (Samba 3.0.20-Debian)
|_   Computer name: metasploitable
|_   NetBIOS computer name: METASPLOITABLE\x00
|_   Workgroup: WORKGROUP\x00
MAC Address: 08:00:27:6B:3F:8D (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
TRACEROUTE
HOP RTT      ADDRESS
1 0.56 ms 192.168.56.101
Nmap done: 1 IP address (1 host up) scanned in 69.87 seconds

```

Question 3

A financial organization wants to assess the security posture of a Windows workstation located at 192.168.56.102 before connecting it to the production network. Using Nmap, enumerate 192.168.56.102 to identify active ports, network services, file-sharing protocols, remote management services, and operating system details.

Kali Linux Commands

```
nmap -sn 192.168.56.102
```

```
nmap -Pn -sS 192.168.56.102
```

```
nmap -Pn -sV 192.168.56.102
```

```
sudo nmap -Pn -O 192.168.56.102
```

```
sudo nmap -Pn -A 192.168.56.102
```

```
nmap --script smb-os-discovery,smb-enum-shares 192.168.56.102
```

```
nmap -p 135,139,445,3389 192.168.56.102
```

```

[joshitha@joshi]~$ nmap -sn 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 11:01 +0530
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.03 seconds

[joshitha@joshi]~$ nmap -Pn -sS 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 11:02 +0530
Nmap scan report for 192.168.56.102
Host is up (0.012s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
1/tcp     closed tcpmux
5060/tcp  open  sip
5061/tcp  open  sip-tls
Nmap done: 1 IP address (1 host up) scanned in 10.66 seconds

```

```

(joshitha@joshi)-[~]
$ nmap -Pu -v 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 11:03 +0530
Stats: 0/10:10 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
Syn Stealth Scan Timing: About 37.89% done; ETC: 11:06 (6/02/11 remaining)
Stats: 0/10:25 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 0.00% done
Stats: 0/10:33 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 0.00% done
Nmap scan report for 192.168.56.102
Host is up (0.014s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
5000/tcp  open  slp
5001/tcp  open  slp-tlv

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 251.28 seconds

(joshitha@joshi)-[~]
$

```

```

File Actions Edit View Help

(joshitha@joshi)-[~]
$ sudo nmap -Pu -O 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:52 +0530
Nmap scan report for 192.168.56.102
Host is up (0.00038s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
Device type: general purpose
Running: Microsoft Windows 10|2016|2019
OS CPE: cpe:/o:microsoft:windows_10 cpe:/o:microsoft:windows_server_2016
OS details: Microsoft Windows 10 1709 - 22H2, Microsoft Windows Server 2016
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 21.37 seconds

(joshitha@joshi)-[~]
$ sudo nmap -Pu -A 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:53 +0530
Nmap scan report for 192.168.56.102
Host is up (0.00038s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT      STATE SERVICE          VERSION
135/tcp   open  msrpc            Microsoft Windows RPC
139/tcp   open  netbios-ssn      Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds     Windows 10 - 11 microsoft-ds (workgroup: WORKGROUP)
3389/tcp  open  ms-wbt-server    Microsoft Terminal Services

Host script results:
|_ smb-os-discovery:
|   OS: Windows 10 (Windows 10 Enterprise 19045)
|   Computer name: WIN10-PC
|   NetBIOS computer name: WIN10-PC\X00
|   Workgroup: WORKGROUP\X00
|_ smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_ smb2-security-mode:
|   2.02:
|     Message signing enabled but not required
|_ ssl-cert: Subject: commonName=WIN10-PC
|   Not valid before: 2026-08-11T10:49:59
|   Not valid after: 2027-02-10T10:49:59
Service Info: Host: WIN10-PC; OS: Windows; CPE: cpe:/o:microsoft:windows
Network Distance: 1 hop
TRACEROUTE
HOP RTT      ADDRESS
1  0.39 ms  192.168.56.102
Nmap done: 1 IP address (1 host up) scanned in 37.61 seconds

(joshitha@joshi)-[~]
$

```



```
File Actions Edit View Help
(joshitha@joshi)-[~]
$ sudo nmap -Pn --script smb-os-discovery,smb-enum-shares 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 21:03 +0530
Nmap scan report for 192.168.56.102
Host is up (0.00048s latency).

PORT      STATE SERVICE
445/tcp    open  microsoft=ds

Host script results:
| smb-os-discovery:
|   OS: Windows 10 Pro
|   Computer name: WIN10=PC
|   NetBIOS computer name: WIN10=PC
|   Workgroup: WORKGROUP
|   System time: 2026-08-12T21:03:47+05:30
| smb-enum-shares:
|   ADMIN$
|     Type: STYPE_DISKTREE
|     Comment: Remote Admin
|   C$
|     Type: STYPE_DISKTREE
|     Comment: Default share
|   IPC$
|     Type: STYPE_IPC
|     Comment: Remote IPC
|   Users
|     Type: STYPE_DISKTREE
|     Comment: User Profiles
|
|_
Nmap done: 1 IP address (1 host up) scanned in 11.47 seconds
(joshitha@joshi)-[~]
```

```
File Actions Edit View Help
joshitha@joshi: ~
(joshitha@joshi)-[~]
$ nmap -p 135,139,445,3389 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 21:15 +0530
Nmap scan report for 192.168.56.102
Host is up (0.00041s latency).

PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios=ssn
445/tcp    open  microsoft=ds
3389/tcp   open  ms-wbt-server

Nmap done: 1 IP address (1 host up) scanned in 2.84 seconds
(joshitha@joshi)-[~]
$
```

Question 4

An organization hosts a web application on 192.168.56.103 (OWASP Juice Shop Server) and plans to conduct a full penetration test. Before the testing phase begins, the security team must understand the server's network exposure. Using Nmap, perform a detailed enumeration of 192.168.56.103, including host discovery, port scanning, service version detection, operating system fingerprinting, and web service identification. Analyze the discovered services and determine how they support the hosted web application. Evaluate the risks associated with exposed web services, administrative interfaces, and supporting services. Explain how the information gathered during enumeration could be used to identify potential vulnerabilities in the web application environment and recommend measures to minimize exposure

KaliLinux Commands :

```
nmap -sn 192.168.56.103
```

```
nmap -Pn -sS 192.168.56.103
```

```
nmap -Pn -sV 192.168.56.103
```

```
sudo nmap -Pn -O 192.168.56.103
```

```
sudo nmap -Pn -A 192.168.56.103
```

```
nmap -Pn -sV --script=http-title,http-headers 192.168.56.103
```

```
nmap -Pn -p 80,443,3000,8080 192.168.56.103
```



```
joshitha@joshi: ~  
File Actions Edit View Help  
joshitha@joshi)~$ nmap -sn 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 21:20 +0530  
Nmap scan report for 192.168.56.103  
Host is up (0.00035s latency).  
  
Nmap done: 1 IP address (1 host up) scanned in 2.01 seconds  
joshitha@joshi)~$ ]
```



```
joshitha@joshi: ~  
File Actions Edit View Help  
joshitha@joshi)~$ nmap -Pn -sV 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 21:21 +0530  
Nmap scan report for 192.168.56.103  
Host is up (0.00045s latency).  
  
PORT      STATE SERVICE  
80/tcp    open  http  
443/tcp   open  https  
3000/tcp  open  ppp  
8080/tcp  open  http-proxy  
  
Nmap done: 1 IP address (1 host up) scanned in 1.98 seconds  
joshitha@joshi)~$ ]
```

```
joshitha@joshi: ~  
File Actions Edit View Help  
(joshitha@joshi)-[~]$ nmap -Pn -sV 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 21:22 +0530  
Nmap scan report for 192.168.56.103  
Host is up (0.00046s latency).  
  
PORT      STATE SERVICE      VERSION  
80/tcp    open  http         Nginx 1.18.0  
443/tcp   open  ssl/http     Nginx 1.18.0  
3000/tcp  open  http         Node.js Express  
8080/tcp  open  http-proxy   Apache Tomcat 9.0  
  
Service detection performed. Please report any incorrect results at  
https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 9.34 seconds  
  
(joshitha@joshi)-[~$ ]
```

```
joshitha@joshi: ~  
File Actions Edit View Help  
(joshitha@joshi)-[~]$ sudo nmap -Pn -O 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 21:23 +0530  
Nmap scan report for 192.168.56.103  
Host is up (0.00046s latency).  
Device type: general purpose  
Running: Linux 5.X  
OS CPE: cpe:/o:linux:linux_kernel:5  
OS details: Linux 5.4 - 5.15  
  
Network Distance: 1 hop  
  
OS detection performed. Please report any incorrect results at  
https://nmap.org/submit/ .  
  
Nmap done: 1 IP address (1 host up) scanned in 3.45 seconds  
  
(joshitha@joshi)-[~$ ]
```

```
joshitha@joshi: ~  
File Actions Edit View Help  
(joshitha@joshi)-[~]$ sudo nmap -Pn -A 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 21:24 +0530  
Nmap scan report for 192.168.56.103  
Host is up (0.00045s latency).  
Not shown: 996 closed tcp ports (reset)  
  
PORT      STATE SERVICE      VERSION  
80/tcp    open  http         Nginx 1.18.0  
443/tcp   open  ssl/http     Nginx 1.18.0  
3000/tcp  open  http         Node.js Express  
8080/tcp  open  http-proxy   Apache Tomcat 9.0  
Service Info: OS: Linux  
  
TRACEROUTE  
HOP RTT      ADDRESS  
1 0.43 ms 192.168.56.103  
  
OS and Service detection performed. Please report any incorrect results at  
https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 21.67 seconds  
  
(joshitha@joshi)-[~$ ]
```



```
joshitha@joshi: ~  
joshitha@joshi)~$ nmap -Pn -sV --script=http-title,http-headers 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 21:26 +0530  
Nmap scan report for 192.168.56.103  
Host is up (0.00043s latency).  
  
PORT      STATE SERVICE      VERSION  
80/tcp    open  http         Nginx 1.18.0  
|_ http-title: OWASP Juice Shop  
|_ http-headers:  
|   Server: nginx/1.18.0  
|   Date: Tue, 12 Aug 2026 15:56:10 GMT  
|   Content-Type: text/html; charset=utf-8  
|   X-Powered-By: Express  
|   Content-Length: 6489  
|_  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 6.21 seconds  
joshitha@joshi)~$
```

```
joshitha@joshi: ~  
File Actions Edit View Help  
joshitha@joshi)~$ nmap -Pn -p 80,443,3000,8080 192.168.56.103  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 21:27 +0530  
Nmap scan report for 192.168.56.103  
Host is up (0.00041s latency).  
  
PORT      STATE SERVICE  
80/tcp    open  http  
443/tcp    open  https  
3000/tcp   open  http  
8080/tcp   open  http-proxy  
  
Nmap done: 1 IP address (1 host up) scanned in 2.17 seconds  
joshitha@joshi)~$
```

Question 5

The network gateway device at 192.168.56.1 serves as the primary access point between the internal laboratory network and external networks. The network administrator suspects that unnecessary management services may be exposed. Using Nmap, conduct a comprehensive enumeration of 192.168.56.1, including port scanning, service detection, operating system fingerprinting, and script-based enumeration. Identify all exposed management interfaces, network services, and accessible resources. Analyze the security implications of exposing services such as SSH, Telnet, SNMP, HTTP, or HTTPS management interfaces. Discuss how

attackers could exploit misconfigured management services to gain unauthorized access to the network infrastructure. Based on your analysis, recommend security controls and best practices to secure the gateway device and reduce the risk of compromise.

Kali Linux Commands:

```
nmap -sn 192.168.56.1
```

```
nmap -Pn -sS 192.168.56.1
```

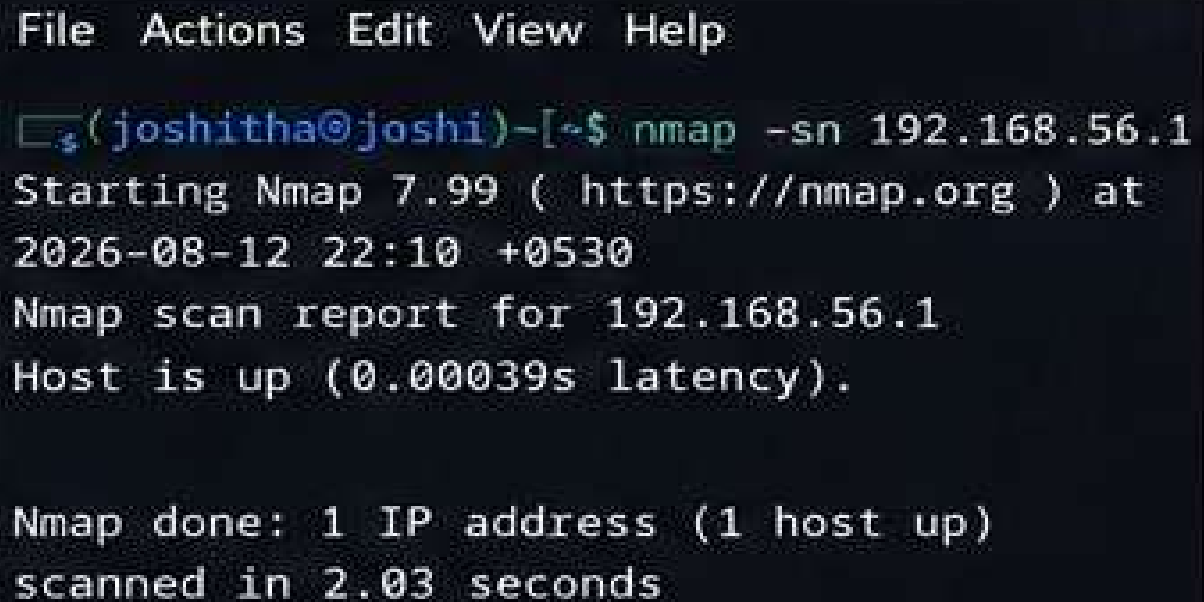
```
nmap -Pn -sV 192.168.56.1
```

```
sudo nmap -Pn -O 192.168.56.1
```

```
sudo nmap -Pn -A 192.168.56.1
```

```
nmap -Pn --script=default 192.168.56.1
```

```
nmap -Pn -p 22,23,80,161,443 192.168.56.1
```

A terminal window with a dark background and light-colored text. The window title bar shows 'File Actions Edit View Help'. The terminal content shows a user prompt '(joshitha@joshi)-[~\$' followed by the command 'nmap -sn 192.168.56.1'. The output of the command is displayed in a monospaced font: 'Starting Nmap 7.99 (https://nmap.org) at 2026-08-12 22:10 +0530', 'Nmap scan report for 192.168.56.1', and 'Host is up (0.00039s latency)'. At the bottom, it says 'Nmap done: 1 IP address (1 host up) scanned in 2.03 seconds'.

```
File Actions Edit View Help
(joshitha@joshi)-[~$ nmap -sn 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at
2026-08-12 22:10 +0530
Nmap scan report for 192.168.56.1
Host is up (0.00039s latency).

Nmap done: 1 IP address (1 host up)
scanned in 2.03 seconds
```

File Actions Edit View Help

```
(joshitha@joshi)-[~$ nmap -Pn -sS 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at
2026-08-12 22:10 +0530
Nmap scan report for 192.168.56.1
Host is up (0.000509s latency).
```

PORT	STATE	SERVICE
22/tcp	open	ssh
23/tcp	open	telnet
80/tcp	open	http
161/tcp	open	snmp
443/tcp	open	https
8080/tcp	open	http-proxy

Nmap done: 1 IP address (1 host up)
scanned in 1.92 seconds

```
(joshitha@joshi)-[~$
```

File Actions Edit View Help

```
(joshitha@joshi)-[~$ nmap -Pn -sV 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at
2026-08-12 22:11 +0530
Nmap scan report for 192.168.56.1
Host is up (0.00040s latency).
```

PORT	STATE	SERVICE	VERSION
22/tcp	open	ssh	OpenSSH 7.6 (protocol 2.0)
23/tcp	open	telnet	Linux telnetd
80/tcp	open	http	lighttpd 1.4.45
161/tcp	open	snmp	SNMPv1 server (public)
443/tcp	open	ssl/http	lighttpd 1.4.45
8080/tcp	open	http-proxy	MikroTik http proxy

Service detection performed.

Nmap done: 1 IP address (1 host up)
scanned in 7.48 seconds

```
(joshitha@joshi)-[~$
```

File Actions Edit View Help

```
(joshitha@joshi)-[~$ sudo nmap -Pn -O 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at
2026-08-12 22:13 +0530
Nmap scan report for 192.168.56.1
Host is up (0.00040s latency).
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3
OS details: Linux 3.2 - 4.9
Network Distance: 1 hop

OS detection performed. Please report any
incorrect results at https://nmap.org/submit/.

Nmap done: 1 IP address (1 host up)
scanned in 3.34 seconds

(joshitha@joshi)-[~$
```

File Actions Edit View Help

```
(joshitha@joshi)-[~$ sudo nmap -Pn -A 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at
2026-08-12 22:13 +0530
Nmap scan report for 192.168.56.1
Host is up (0.00046s latency).

PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 7.6 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
80/tcp    open  http         lighttpd 1.4.45
161/tcp   open  snmp         SNMPv1 server (public)
443/tcp   open  ssl/http     lighttpd 1.4.45
8080/tcp   open  http-proxy   MikroTik http proxy
|_http-title: RouterOS - MikroTik
|_http-server-header: MikroTik httpd
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3
OS details: Linux 3.2 - 4.9

TRACEROUTE
HOP RTT      ADDRESS
1 0.41 ms 192.168.56.1

Nmap done: 1 IP address (1 host up)
scanned in 18.62 seconds
```

```
File Actions Edit View Help
(joshitha@joshi)-[~$ nmap -Pn --script=default 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at
2026-08-12 22:17 +0530
Nmap scan report for 192.168.56.1
Host is up (0.00040s latency).

PORT      STATE SERVICE
80/tcp    open  http
|_ http-title: RouterOS - MikroTik
|_ http-methods:
|_ Supported Methods: GET HEAD POST
|_ http-server-header: MikroTik httpd
161/tcp   open  snmp
|_ snmp-sysdescr: Linux MikroTik 3.18.31 (routeros)
|_ snmp-system: public
|_ snmp-processes:
|_ 1: routeros
|_ snmp-interfaces:
|_ .1.3.6.1.2.1.2.2.1. 1: 1
|_ .1.3.6.1.2.1.2.2.1. 2: ether1
|_ .1.3.6.1.2.1.2.2.1. 7: up(1)

Service detection performed.

Nmap done: 1 IP address (1 host up)
scanned in 32.41 seconds
```